

Exam Date & Time: 17-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: CSE 3121/CSE_3121 PRINCIPLES OF CRYPTOGRAPHY

(SPL: COMPUTER SCIENCE & ENGINEERING)

PRINCIPLES OF CRYPTOGRAPHY [CSE 3121]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

1A) Analyze how the AES key schedule design enhances overall security and ensures effective diffusion throughout the encryption process. (5)

1B) Given the plaintext "CRYPTOGRAPHY" and the key matrix:

$$K = \begin{pmatrix} 3 & 4 & 5 \\ 5 & 4 & 2 \\ 2 & 1 & 4 \end{pmatrix} \quad (3)$$

Encrypt the plaintext using the Hill Cipher.

1C) Find the following:
i) $\phi(512)$ (Using Euler's Totient function) (2)
ii) $7^{301} \bmod 11$ (Using Fermat theorem)

2A) Analyze how the 1024-bit message block is transformed into an 80-word input sequence during the SHA-512 message schedule process with a neat diagram. (4)

2B) Analyze the working of the RC4 stream cipher algorithm with a suitable example. Explain how the design of RC4 contributes to stream generation and discuss potential weaknesses arising from its internal structure. (3)

2C) Illustrate how diffusion and confusion operate in the DES block cipher, and show with examples how they contribute to making the ciphertext more secure. (3)

3A) Determine the value of x for the following linear set of congruences using Chinese remainder Theorem (CRT):

$$\begin{aligned} x &\equiv 2 \pmod{7} \\ x &\equiv 5 \pmod{9} \\ x &\equiv 7 \pmod{11} \\ x &\equiv 3 \pmod{16} \end{aligned} \quad (4)$$

3B) In a public-key system using **RSA**, you intercept the ciphertext $C = 20$ sent to a user whose public key is $e = 13$, $n = 77$. Evaluate the plaintext M . Evaluate the process by calculating the private key and decrypting the ciphertext to recover the plaintext message. Show the modular exponential steps. (4)

3C) Explain any four design principles of a block cipher. (2)

4A) Justify the Interchange of Add Round Key Transformation and Inverse Mix Columns Transformation of Advanced Encryption Standard (AES) encryption algorithm to obtain the Equivalent Inverse Cipher with relevant derivations. (3)

4B) (4)

Alice and Bob agree to use the **Diffie–Hellman key exchange** to establish a shared secret key for secure communication. They choose a common prime modulus $q = 23$ and a primitive root $a = 5$. If Alice's private key is $X_A = 6$ and Bob's private key is $X_B = 15$, compute their corresponding public keys Y_A and Y_B . Further, determine the **shared secret key** generated between Alice and Bob and briefly analyse how the use of private keys ensures confidentiality in the exchange.

- 4C) Analyze techniques for generating random and pseudorandom numbers and the key characteristics required to test randomness. (3)
- 5A) Alice and Bob share a secret key K. Alice sends a message M with its MAC computed. When B receives M and recomputes MAC, the tags do not match. Analyse the conclusion that B should draw, and which security objective has likely been violated? Explain with equations. (2)
- 5B) Analyse how a man-in-the-middle attack operates in the Diffie-Hellman key exchange by intercepting and modifying the exchanged public keys between Alice and Bob. Explain, with reasoning, why the attacker requires two different key pairs to successfully establish independent shared keys with both parties. (4)
- 5C) Explain the rationales of using MACs instead of just using symmetric encryption for authentication. (4)

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Event Name	Subject
058-END SEMSTER-ODD-2025	CSE_3121:Principles of Cryptography

#	Course Outcome	Description	Marks
1	1	CO 1	3
2	2	CO 2	13
3	3	CO 3	12
4	4	CO 4	12
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name	Subject
058-END SEMSTER-ODD-2025	CSE_3121:Principles of Cryptography

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	2(4%)
3	Application	4	11(22%)
4	Analysis	6	23(46%)
5	Create	4	14(28%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)